

Multiple Choice

1. **Answer:** A

Options: A) No, because cancer is not contagious.; B) No, because cancer is not always fatal.; C) Yes, because millions of people have cancer.; D) Yes, because people all over the world have cancer.

Note: Cancer is not classified as a pandemic because it does not spread from person to person, as it is not contagious like infectious diseases.

2. **Answer:** A

Options: A) lowering parking fees; B) building larger parking lots; C) decreasing the cost of gasoline; D) lowering the cost of bus and subway fares

Note: Lowering parking fees can encourage energy conservation by incentivizing individuals to use their vehicles more efficiently or switch to public transportation, thereby reducing overall energy consumption associated with driving.

3. **Answer:** B

Options: A) lighting a match; B) breaking a glass; C) burning of gasoline; D) rusting of iron

Note: Breaking a glass is considered a physical change because it alters the form of the glass without changing its chemical composition.

4. **Answer:** D

Options: A) increase amount of heating; B) increase surface area of walls; C) increase number of windows; D) increase thickness of insulation

Note: Increasing the thickness of insulation is the best solution to reduce heat loss in a building during winter, as thicker insulation provides better thermal resistance, minimizing the transfer of heat from the interior to the colder outside environment.

5. **Answer:** D

Options: A) arteries; B) capillaries; C) valves; D) veins

Note: Veins are responsible for carrying de-oxygenated blood from the body back to the heart, making them essential for the circulatory system's function.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Both an apple seed and a fertilized egg contain an embryo as they represent the early developmental stages of plants and animals, respectively, where the embryo is the initial form of the organism that will grow and develop into a mature individual.

7. **Answer:** False

Options: A) True; B) False

Note: Sexual reproduction involves the combination of genetic material from two parents, resulting in offspring that have a unique mix of traits and variations rather than being identical to either parent.

8. **Answer:** False

Options: A) True; B) False

Note: Antibodies do not produce proteins that engulf and digest viruses; instead, they are proteins themselves that bind to viruses to neutralize them and mark them for destruction by other immune cells.

9. **Answer:** True

Options: A) True; B) False

Note: Water molecules form a crystalline structure when frozen, which causes them to be arranged further apart compared to their arrangement in liquid form, resulting in ice being less dense than liquid water and allowing it to float.

10. **Answer:** True

Options: A) True; B) False

Note: The increase in the number of homes built leads to habitat destruction and fragmentation, which negatively impacts the Miami Blue butterfly's food sources and breeding grounds, contributing to its population decline.

Long Form Response

11. **Answer:** Electromagnetic energy can be harnessed as the sole power source for robots by utilizing principles of electromagnetism, which involves the relationship between electricity and magnetism. For instance, solar panels convert sunlight into electrical energy, which can power robotic systems. Additionally, electromagnetic induction can generate electricity through the movement of a conductor within a magnetic field, enabling robots to operate without traditional batteries. Potential applications in engineering include solar-powered drones, automated vehicles, and robots used in remote or off-grid environments, showcasing the versatility of using electromagnetic energy in modern robotics. This approach not only promotes sustainability but also enhances the efficiency of robotic systems.

Note: correct because it effectively connects the principles of electromagnetism, such as solar energy conversion and electromagnetic induction, to the practical application of powering robots without reliance on conventional batteries.

12. **Answer:** Las Vegas has several renewable energy resources, notably hydroelectric power, which harnesses the energy from flowing water. The nearby Hoover Dam is a significant source of hydroelectricity, providing clean energy while also supporting local water supply and recreation. However, hydroelectric power can have environmental impacts, such as altering fish habitats and affecting water quality. To utilize these

resources sustainably, it is important to balance energy production with environmental protection, such as implementing fish ladders to support aquatic life and maintaining natural river flows. By carefully managing these impacts, Las Vegas can continue to benefit from renewable energy while preserving its ecosystems.

Note: correct because it identifies hydroelectric power as a key renewable energy resource in Las Vegas, discusses both its benefits and environmental impacts, and emphasizes the need for sustainable practices in its utilization.

End of Answer Key